

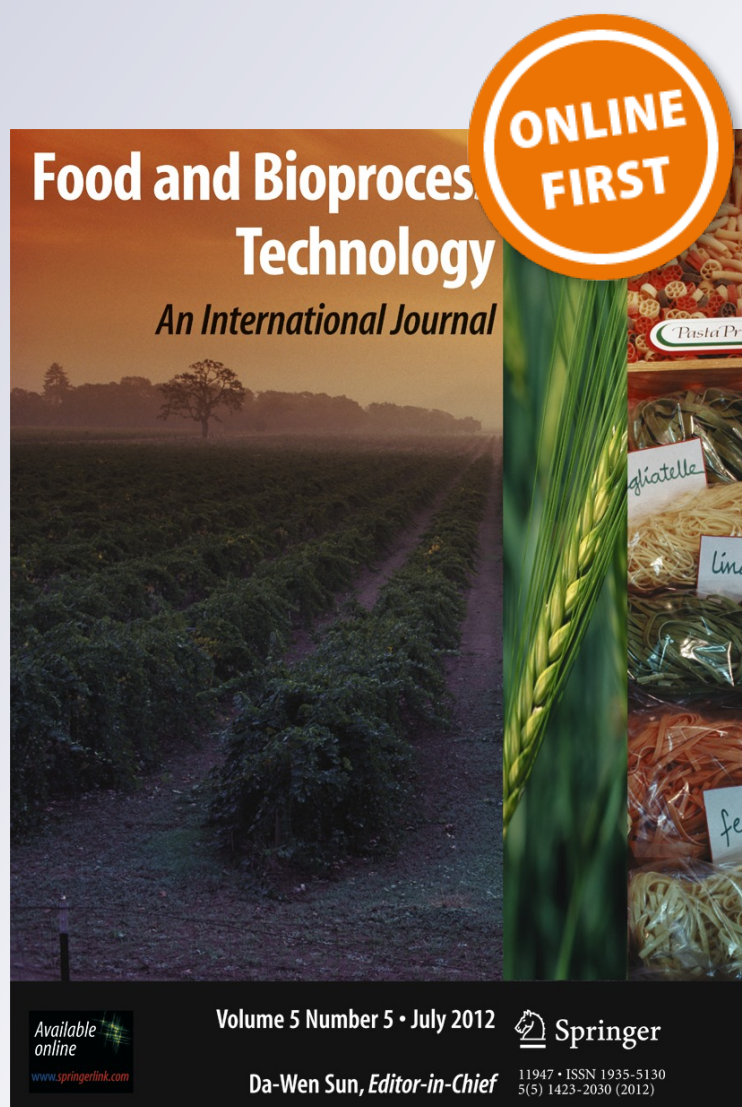
A Review on Agri-food Supply Chain Traceability by Means of RFID Technology

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Abstract Radio Frequency Identification (RFID) is a technology which provides appealing opportunities to improve the management of information flow within the supply chain and security in the agri-food sector. Nowadays, food safety is considered a major requirement in several countries, in particular, the traceability of food products which is mandatory by law. Thus, technological implementation leading to traceability strengthening in the agri-food sector is crucial. The first aim of this review is to analyze the current developments in RFID technology in the agri-food sector, through an operative framework which organizes the literature and facilitate a quick content analysis identifying future research direction. RFID technology seems to be able to bring great opportunities to this sector; nevertheless, several constraints are slowing its adoption. This survey may provide readers with an exhaustive overview of opportunities and constraints for the wide adoption of RFID. The second aim of this review is to provide an updated analysis on the current developments of RFID technology for different product typologies within the agri-food industry, discussing at the same time its potential in technological and logistical development regarding different sectors of the production/distribution chain. As referenced here, RFID

implementations in the agri-food sector are increasing at a fast rate, and technological advancement follows the applicability opportunities. However, real applications of RFID technologies are still limited because of various technical and economical obstacles which are also discussed.

Keywords Radio Frequency · Infotracking · Supply chain · Food control · Logistic · Wireless sensing

Introduction

Traceability is considered today a crucial factor for the agri-food sector. The specific definition of traceability within the agricultural framework of the agri-food domain as it can be found on the OnTrace (2007), refers to it as “The ability to locate an animal, commodity, food product or ingredient and follow its history in the supply chain forward (from source to consumer) or backward (from consumer to source).”

Within this context, traceability is a mandatory requirement for the agri-food sector in many countries, bringing many benefits such as increasing the security of customers, and so their confidence, and controlling the effects of commodity withdrawal (Gandino et al. 2009; Nambiar 2010). In the Regulation EC No 178/2002 (European Commission 2002), food traceability has a strictly regulated procedure. Food sector business should allow the precise identification of the origin of each food product through all the production and distribution stages.

Traceability policies are getting implemented on the basis of consumers' studies. The consumers rely on traceability as a tool to ensure certain processing aspects and correct strategies (e.g., harvesting processes and safety procedures). Moreover, supply chain improvement, food safety control and accessing foreign market segments are strong incentives for primary producers and processors to participate in programs with traceability features (Trautman et al. 2008).

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Besides being an instrument with a potential capacity to provide added information for all participants along the production and distribution chain, once the protocol is set up it represents an efficient tool for sharing information. Moreover, its real value is represented by the shortened time needed to intervene in the case of a food safety incident to recall an entire stock from the market and individuate the real problem origin. Nevertheless, such systems can reduce transaction and displacement costs.

Radio Frequency Identification (RFID) technology is a flow control technology which allows traceability of goods through all steps of production chain (Kelepouris et al. 2007; Ngai et al. 2008a; Nambiar 2009; Shi et al. 2009). RFID technology uses a few simple and inexpensive components (see Section 2), which in the past decade progressively gained importance in the traceability technological scenario, since it allows (Sarac et al. 2010): (a) omitting the manual control; (b) increasing versatility of operational contexts (e.g., being water-proof, antimagnetic, supporting a wide range of temperatures); (c) versatility of logistic context (permitting a very different range of reading distances); (d) long service life; and finally; (e) miniaturization to a certain extent. Moreover, as data may be encrypted on the electronic label, the storage capacity is big (from few dozen of bits to 32Kbytes) and the stored data can be changed, according to the tag type, giving the possibility to correct eventual errors committed during the information flux or to add information that at the beginning were unknown.

The popularization and versatility of RFID technology will bring revolutionary changes to industry retail and logistics, eventually replacing the ubiquitous Universal Product Code (UPC) identification, mostly known as “bar code” (Michael and McCathie 2005; Singer 2006; Regattieri et al. 2007). Compared with the presently still popular bar code and QR-code technologies (similar to bar codes, readable by a QR code reader; Seino et al. 2004), RFID tagging has many advantages that can help enterprises to increase the speed of the flowing information, incrementing the exchange quantity *per* relative displacement time needed and saving of production and distribution costs. As reviewed by Biedeman (2006) and Shi et al. (2009), there are a series of advantages in using RFID in relation to the more traditional barcode one since an increased automatism in tagging and reading will allow users to: (a) reduce stock and the sales personnel costs (e.g., reduced labour cost of products identification); (b) increment efficiency in stock turnover; and finally, (c) reduced larceny occurrence.

RFID traceability applications in the agri-food industrial environments are mostly directed to the production/distribution chain which is composed of the following subjects (Gandino et al. 2009): (a) producers of raw materials which sell them to the manufacturing enterprises; (b) manufacturing enterprises that transform the received products into

marketable items; (c) distributors that move alimentary commodities toward their final commercialization (normally, in a distribution center commodities are not treated, they are only stored under determined conditions and then shipped to their destination; Menesatti et al. 2012); (d), retailers that sell alimentary commodities directly to customers; and finally, (e) transporters should also be added to this chain, because they are responsible for the delivery of alimentary products from a production chain subject to another. The components of this technology are becoming smaller and smaller, less expensive and more effective (Sarac et al. 2010). Thus, applications of RFID in supply chain are increasing. Bagchi et al. (2007) reported the prediction of RFID growth from \$1 billion in 2003 to \$4 billion in 2008 to \$20 billion in 2013.

Within this context, the aim of this review is to produce an updated analysis on the current developments of RFID technology for different product typologies of the present agri-food industry, discussing at the same time its potential technological and logistical development within different sectors of the production/distribution chain.

RFID Devices

RFID architecture is composed by the following components (Roberts 2006; Aguzzi et al. 2011): (1) an RFID tag, which can be directly applied on the displacing good, (2) an interrogator (or antenna), as a device (i.e., the reader) that gathers information from the tag (ID or data stored), and (3) a database system used to store the information gained through the interrogation routines carried out by the antenna.

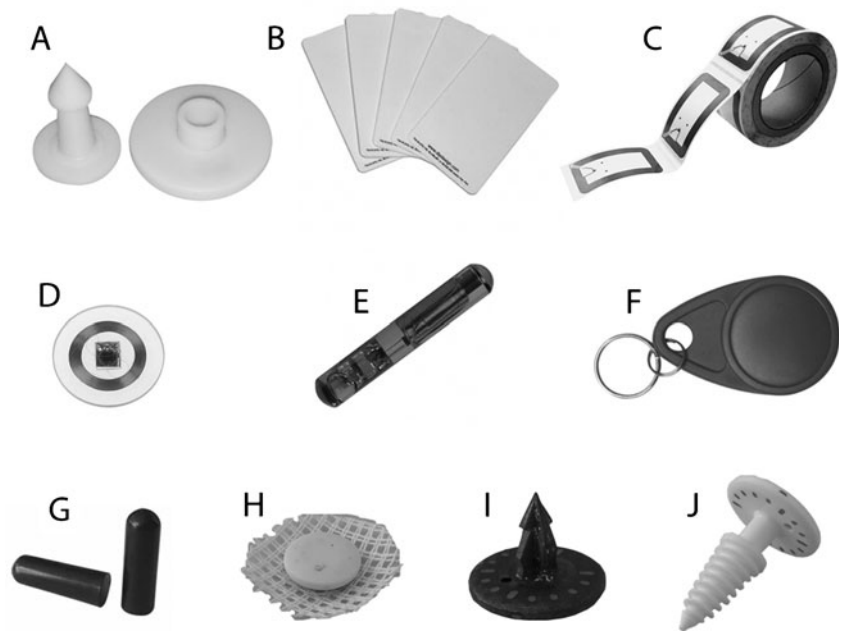
An RFID interrogator (either defined by the synonymous “reader” or “antenna”) is a device that recognizes the ID information of the tag upon backscattered communication (see below). Readers can either be portable on handheld terminals or fixed as devices that can be positioned in different strategic places of the production/distribution chain or reading site.

RFID tags, also known as transponders, are usually small pieces of material, typically comprising three components: an antenna, a microchip unit containing memory storage and an encapsulating material of different shapes (Fig. 1). Depending on their application, tags can be embedded in glass or epoxy resins or other materials.

The tags have an identification code (i.e., the Electronic Product Code) commonly of 96 bits recorded at the time of manufacture. Tags can be classified in terms of data access in read-only or read-write tags. These terms refer to whether or not the information stored in the tag can be changed or erased. In a read-only tag the data cannot be modified or appended but it may be read multiple times and they are

Fig. 1 Examples of RFID tags.

a Ear tag suited for use with cattle, sheep, pigs and other livestock for RFID management and scientific testing applications; **b** card for use in access control systems with data memory; **c** embedded within normal barcode labels for added extra security or unattended reading and sorting operations; **d** small LH frequency clear RFID tag, suitable for use with animals in a collar, etc., for ID and tracking; **e** injectable tag; **f** key applications; **g** capable of withstanding harsh environments embedded with material like rubbers and furniture; **h** cheese tag; **i, j** Ham, meat and similar products



often used in simple tracking applications. Differently, in a read–write tag the stored information can be modified or written several times but also can be write-protected. Tags of this latter category usually offer greater functionality, as they are can be reused several times or used for applications where data regarding production, manipulation and traceability of the good are required (AAVV 2006). Because of this, read–write tags are usually more expensive than read-only tags. EEPROM is the memory typical used for RFID tags and is usually divided into blocks or pages where user-memory and the ID (not rewritable) are stored and mapped. Memory capacity varies from few dozen of bits to 32 KB, usually enough in the vast majority of applications. There is no existing standardized format of data, so the final user can use the outline and data structure suitable for their applications (these could be product or environmental measurements, time-stamps and locations of the good in the supply chain and so on) or include cryptographic methods to restrict their access and improve safety (Microchip Technology Inc 2004). Ideally, the tag itself should carry minimal valuable data, or else include a trusted security mechanism for data protection.

Other specifications of the tag apart from their applicability on goods and memory are related to their activity or passivity when transmitting the information (Aguzzi et al. 2011). Passive tags emit their information only when powered by the energizing field of the interrogator (i.e., the antenna; see below). This means that the tag is only powered when the bearing object enters in the range of action of the interrogator as well as using this radio wave to carry the data, and the tag is able to convert it into power. The tag then uses a technique called backscatter to reply to the interrogator. This does not involve a transmitter on the tag,

but is a means of “reflecting” the carrier wave and putting a signal into that reflection.

Conversely, active tags are those which bear a battery and are able to transmit independently from the powering field of interrogators. These tags are just like passive tags since they use backscatter principle to convey their ID information but they have a battery assisted so they can provide the power source to operate the chip in an autonomous fashion. Transmitting autonomy conveys great advantages to RFID technology, because transmission can cover greater distances from the interrogator greater than several tens of meters. Active tags have not only a battery, but also some form of transmitter on the tag. The antenna in a tag is the physical interface for the RF to be received and transmitted. Its construction varies depending on the tag itself and the frequency it operates on. Low frequency tags often use coils of wire, whereas high frequency tags are usually printed with conducting inks. Depending on environmental conditions and maintenance, barcode read rates often decline to less than 90 % over time. In most environments, RFID can achieve 99.5 % to 100 % first-pass read rates (Texas Instruments 2006). RFID does not require line-of-sight with the reader, so tags can be located inside the products or inside containers.

RFID falls within the category of smart tags in relation to their capacity of data storage and real-time traceability. In relation to the technical specification of RFID tagging technology as described above, some evident operative advantage can be listed in relation to the already well-established bar code identification procedures. First of all, an elevated number of tags can be contemporarily read in an independent fashion, from the same interrogator (Aguzzi et al. 2011). Each tag can uniquely identify the object to which

it is attached, even if that object is one of a multitude of identical items. In second instance, tags do not require a direct line of sight for reading and may be read not only through air cut also through hard material (e.g., book covers or packaging layers) including water (Dobkin and Weigand June 2005). Tag ID information is encoded under according to frequency specifications, the data bit/transfer rate, the method of encoding and any other parameters that may be needed. ISO 18000 is the standard for the air interface for item management.

For active tags, the autonomy of the batteries depends of each tag, but normally is of several years. Some models, in order to solve the problems associated with the battery expiration, are able to send to the reader the battery level, this feature permits to the management system to detect those tags that are going to expire so the battery could be change before exhaustion. The most commonly used batteries are lithium and magnesium dioxide, and they have a weight of about 3 g. Nowadays, ultrafine batteries are being applied; they are flexible and are 1 mm thick. Taking advantage of the low current required by the integrated circuit, paper batteries are also used (AAVV 2006).

RFID technology is not only based on the presence of tags and readers but it requires other software and hardware specifications in order to manage the information load through space and time (Sarriá et al. 2009a, b; Costa et al. 2011a). The most important component is the development of specific RFID management software that translates the raw ID data from the tag into information to be related to other parameters of importance associated to the tag bearer, being this a good or a live stock. This information can then be fed into other databases and applications (e.g., inventory management) for further processing. In the case of read-write tags, specific software is also required in order to manage data writing on the tag itself, containing this latter specification to initiate the process of adding or removing data.

Wireless and RFID

Wireless Sensor Network (WSN) represents a network of sensors used together for the collection and transmission of different types of information gathered from the surrounding environment. A WSN is a system comprised of RF transceivers, sensors, microcontrollers and power sources (Wang et al. 2006). A central node collects the information from a group of spatially related sensors and facilitates communication with a backboard managing centre. Sensors are able to exchange data together and with other external users. Specific software applications manage the network system in collecting and processing of large volumes of data (Jain and Vijaygopalan 2010). In this context, the RFID technology can be adapted to WSN operative environment since its

functioning is based on the transmission of the object ID under the form of a unique serial number using radio waves. The RFID system can be integrated into the WSN network by connecting the tag readers to an RF transceiver, which has routing function and can forward information to and from other readers. An Internet application can be therefore used to detect and monitor changes of the physical status of connected objects through sensors and RFID in real time.

Recently, low-cost, low-power, multifunctional sensor nodes in WSN technology have been developed (Fig. 2). Sensor nodes enable environment sensing together with data processing.

Generally, WSN can be a good solution for monitoring with the appropriated routing and network topology (Ruiz-Garcia 2008). The WSN and RFID synergic integration allows a consistent increase of monitoring possibilities. WSN can be based on a wide variety of different monitoring sensors which cannot identify single objects, a potentiality introduced by the RFID itself. According to Ruiz-Garcia et al. (2009), two standard technologies are presently available for WSN: ZigBee (defined the physical and the Medium Access Control layers for low-rate wireless personal area networks) and Bluetooth (developed as a wireless protocol for short-range communication in wireless personal area networks replacing mobile devices).

Different solutions of integration among WSN, RFID and managing software are proposed in the literature. Jedermann et al. (2006, 2009) described the use of intermodal containers. These containers operate based on the principle of artificial intelligence. Zhang and Wang (2006) described three different ways of WSN and RFID architecture integration. The first one is represented by mix RFID tags and WSN sensor nodes within the same area. A station gathers information from tags and sensor nodes at once, hence transmitting it to local host computer or remote server. The second architecture is represented by a “smart node,” which uses different types of sensors, to detect interested physical scenario, reading RFID tags and radio transceiver which

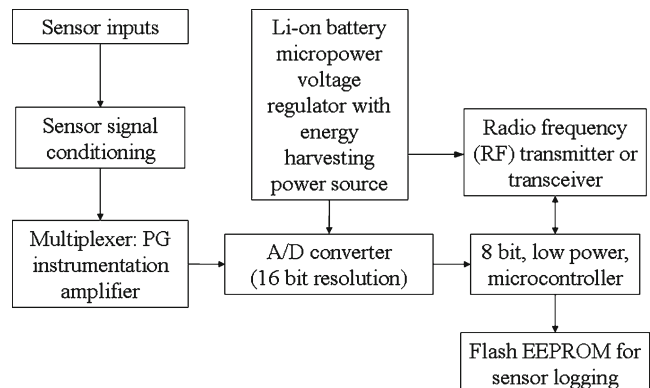


Fig. 2 Example of wireless sensing node (WSN; adapted from Ruiz Garcia, 2008)

transporting sensed data. The third architecture is based on the replacement of RFID tags (which are of active type) by Xbow motes. The active tag is similar to the Xbow mote, but they are not exactly sensor network nodes because they communicate in centralized mode and cannot cooperate with each other through a formed ad hoc network (Ruiz-Garcia 2008). Ruiz-Garcia et al. (2007) presented a monitoring system for good distribution based on network-adapted containers. WSN and RFID distributed devices can be placed in vehicles in order to monitor the environmental conditioning at all moments of the transporting. Anastasi et al. (2009) implemented a system for monitoring the productive cycle of Sicilian winery, where sensor nodes are deployed both in the field and in the cellars where the wine aging occurs. Pereira et al. (2008) also described a combined WSN and RFID system architecture but in relation to tracking and monitoring of animals status. Special wireless sensor nodes are deployed in order to read mixed active (long range) and passive (short range) RFID tags carried by animals, at the same time that they read a diversified set of environmental variables through a series of sensors (e.g., temperature and humidity) that are implemented.

Among these variables, temperature is crucial for the maintenance of products' freshness and appearance. The fast development of IT tools is allowing a fast implementation of cold chain traceability systems within the industry (Hsu et al. 2008). In recent years, many solutions applying different and emerging technologies have been proposed to improve the traceability of fresh products, including seafood. RFID can be used to automate identification of the items at all steps of elaboration, based on the storage capacity of small tags inserted onto the goods (Grabacki et al. 2007). This technology is currently used to construct traceability systems that can be coupled with different steps of the cold chain and distribution (Kumar et al. 2009; Karlsen et al. 2011; Trebar et al. 2011). RFID is progressively substituting bar codes in managing different steps of the cold chain processing, in relation to the verification of legislation conformity, processing, stoking, and transportation monitoring (Aung et al. 2011). In relation to more traditionally employed temperature data loggers, RFID tags present important advantages based on their resistance to humidity and mechanical resistance and lack of human intervention during data retrieval, which can be done remotely (Abad et al. 2009). RFID data loggers can be integrated into a network of sensors which can be mounted on different surfaces, being suitable to read temperature levels and hence capable of efficient monitoring in very different environments such as warehouses and containers (Jedermann et al. 2009). Therefore, the architecture of this detection system can be designed following the industry requirements, being adaptable to both small and medium enterprise (Chen 2008).

Despite the different examples introduced above, the development of an efficient and integrated WSN and RFID technology is still at a early stage and applications to agriculture and food industry are still rare. In the coming years, the lowering cost of WSN and RFID will provide the opportunity to track and trace not only large and expensive products, but small and cheap ones, creating a new generation of sensor networks (Meyer et al. 2009). Products and environmental information that may influence their state will be monitored and traced from the field to the industry, throughout all production–distribution steps.

Infotracking

The integration of innovative systems such as RFID tracking with geostatistics for optimizing supply chain and logistics decisions (Menesatti et al. 2012) may offer competitive great opportunities for the entire supply chain. In order to allow the maximum level of management in the flow of information trough that chain, the focus becomes the traceability of products that are based on innovative web-based systems. Web-based tracing system aims to improve products quality by increasing the level of information transparency for the consumer. This objective occurs by collecting a set of scientific and productive information which follow the product shelf life from producer to consumer, providing a specific web-based tool for each category. The subject categories involved are divided into manufacturers, wholesalers, resellers, retailers and finally the consumers. The implemented web software is structured to provide various services to all the categories thanks to the Application Programming Interface (API), which allows the different subjects of the production/supply chain to implement the acquisition and/or writing system following their needs and available technologies and ensuring uniformity of data to consult or send. Infotracking systems can also provide a reference web interface to access the product info card displaying all information and data released as feedback by the manufacturer, wholesaler, reseller, retailer and consumer (Papetti et al. 2012).

Each element contributes separately, according to a certain level of membership to the data collection in relation to the product at each stage of the production/supply chain. All the collected data can be entered into a centralized database. Computer companies, specialized in Data Processing Centre (DPC), manage large databases by providing web hosting and backup services. This management structure guarantees the uniqueness and the centrality of the acquired data maintaining a controlled access for each system part in order to ensure their integrity in relation to standards procedures complying with the current legislation (e.g., ISO/IEC 27001: 2005).

In particular, some of the different subjects in the production/supply chain involved in the infotracking process

can be listed according to their level of interaction in data collection, processing and storage.

Manufacturer

The product is identified by RFID technology (personal computer and RFID tags) and its quality information, e.g., those regarding the production methods or transport timing, is stored into the centralized database through a web application.

Wholesalers, Resellers and Retailers

The categories involved in this phase can monitor the supply chain of each product through the centralized database and improve their tracing by adding quality information into the system through a web application.

Consumers

The consumers can control the supply chain of each product using the RFID readers provided by the resellers and/or retailers, by the web (browser personal computer) and smartphone applications (APP) inserting the RFID tag code. Also, the consumers can improve the product tracing by adding feedback information on the quality of the product into the system through a web service as blog and forum.

Research Institutions and Statistics

Through the centralized database, all research institutions can use the quality information collected by categories, for statistics and marketing scopes.

Applications on Agri-food Products

The production and large geographic-scale distribution of a diversified set of goods such as horticultural, meat, fishery, dairy, bakery, beverages and other products are proposing logistic and safety challenges which require an implementation of the management technology (Opara and Mazaud 2001). RFID applications have been progressively developed in the past decade in different fields of agri-food production in relation to good traceability through all stages of the production/distribution chain (Ruiz-Garcia and Lunadei 2011). In some cases, RFID applications allowed an increased production efficiency; a group of examples are presented below in relation to development attempt, trials and obtained results. Table 1 lists papers on RFID applications on agri-food products.

Horticultural Products

In the horticultural sector, technological innovation in logistics is crucial from an economic point of view, as they are related to important aspects such as product appearance (Costa et al. 2011b) and freshness (Luvisi et al. 2010a; Menesatti et al. 2012). For this reason, in key areas such as floriculture, the use of RFID technology is progressively gaining importance for several reasons (Yasothkumar 2010): (a) the easier way of tracking the product from its production to its distribution within large storage/selling areas; (b) the best way to visualize, identify and therefore more effectively rectify temperature-related problems at the distribution; (c) the tags are easily programmed to retain, gather and display a very large amount of information usable to improve business operations and derived benefits.

Other horticultural food sectors gained additional benefits from the implementation of RFID technology in comparison with non-food products due to human health implication. This is the case for producers and marketers of fresh fruits and vegetables which can use products traceability to efficiently withdraw their traded good in cases of sanitary crises (e.g., Dole Food Company was able to start a preventive withdrawal of packaged salads that were suspected of *E. coli* bacteria contamination before any infected consumer was reported to get ill; Uldrich 2007).

The main advantage of adopting RFID technology in the traceability of goods, in the supply chain management, is related to quality especially when processing/storage temperature comes into play. Gandino et al. (2009) proposed and tested the use of RFID traceability systems in fruit warehouses. That technology was developed by Vergara et al. (2007), who integrated into RFID readers micro-machined metal oxide gas sensors and showed as they can monitor climacteric conditions during transport and vending. The sensors can be set to work as an alarm, being able to work only within a certain temperature range. This operative limitation allows to users efficiently check the conservation stage of commonly traded fruits such as apples. Moreover, if these sensors are operated under an optimised temperature-modulation mode, accurate calibration models for the storage of each variety can be implemented. In a later study by Ampatzidis et al. (2009), two methods based on the RFID technology were proposed for matching bins containing harvested fruits to overcome the limitations of existing yield mapping systems for manual fresh fruit harvesting. In the first method, a long-range RFID reader and a Global Positioning System in differential mode (DGPS) were assembled together on an orchard tractor, while a passive RFID tag was placed onto the bins. In the second method, the DGPS was not used and RFID tags were instead attached to individual trees as well as on bins. Also, Hertog et

Table 1 Lists of papers on RFID applications and technologies on agri-food products

Food	Product	Technology	Ref.
Horticultural products	Fruits general	RFID reader with on-board micromachined metal oxide sensors	Gandino et al. 2009; Ruiz-Garcia et al. 2007; Chunxia et al. 2009; Yang and Wang 2012
		RFID radar-reader, and high gain patch antennas and RFID tag	Vergara et al. 2007
		RFID tags comparison with barcodes	Anpatzidis et al. 2009
		Four kinds of RFID temperature sensors	Anpatzidis and Vougioukas 2009
	Pineapple		Anador et al. 2009
	Tomato		Hertog et al. 2008; Milczarek and McCarthy 2011
	Fresh cut salads	RFID HF 13.56 MHz with unique identification data and rewritable memory	Antonucci et al. 2009
	Flowers general	RFID microchip internal implants	Yasothkumar 2010
	Grapewine	RFID microchip internal implants	Bandinelli et al. 2009; Luvisi et al. 2010b, c
	Rose	RFID tags comparison with barcodes	Luvisi et al. 2010a
Meat products	Meat General		Tomeš et al. 2009
	Beef	Sensor bolts with embedded RFID tags, RFID readers and network software	Kerry et al. 2006; Shougang et al. 2010
	Pig	RFID tags and barcodes	Collins 2004
	General	Custom RFID tag for carcass	Kong et al. 2009
Dairy products	Cheese	Personal Computer vs. PocketPC	Luo et al. 2011
		Two typologies of tags tested	Tang 2009; Zhang et al. 2010
	Milk		Pérez-Aloe et al. 2007
	Seafood general		Regatieri et al. 2007; Varese et al. 2008
Fishery products	Fish		Papetti et al. 2012
			Xin and Stone 2008; Rizzi et al. 2011; Cai and Liang 2011
			Thompson et al. 2005; Sioen et al. 2007; Schröder 2008
			Chen 2008; Trebar et al. 2011
		Sensors controlled by the PLC and web-based system	Hsu et al. 2008
		Smart tag which integrates light, temperature and humidity sensors, a microcontroller, a memory chip, low power electronics and an antenna for RFID communications	Abad et al. 2009
		Permittivity at UHF RFID frequencies	McCarthy et al. 2009
		915 MHz RFID system	Laniel and Émond 2010
		RFID with temperature sensors	Liu et al. 2010; Tingman et al. 2010
		RFID HF 13.56 MHz with unique identification data and rewritable memory	Costa et al. 2011b
Bakery products	Lobster		Sanja et al. 2009b
	Smoked seafood		Grabacki et al. 2007
	Bakery general		Smith et al. 2004
	Bread	Tags at different frequencies (915 and 433 MHz)	Laniel et al. 2011
Beverages	Beverages general	Tag antenna based sensing principle	Bhattacharyya et al. 2010
	Beer		Swedberg 2006
	Wine	Personal digital assistant (PDA) with a key mechanism involving both the passive RFID internal memory and the unique RFID identifier	Bernardi et al. 2008
		WSN-based system	Anastasi et al. 2009
Other food products	Egg	Data recorder with RFID features	Singh et al. 2007
	Sushi	Architectural framework of an RFID system on a conveyor belt	Ngai et al. 2008b
	Pasta, coffee, milk, olive oil		Rizzi et al. 2011

al. (2008) monitored Belgian tomatoes from growers to retail chain, using RFID labels with integrated temperature sensors. The monitored temperature scenarios were used in simulations to further validate the quality change model. Different batches of product were identified and characterised at the growers' farm, and finally the product was monitored during transport with RFID labels and recovered after going through the marketing chains. Similar studies were performed by Amador et al. (2009), who used RFID technologies for the temperature mapping of the pineapple supply chain by comparing the performance of temperature tags, with and without probes, versus other more conventional methods (i.e., HOBO temperature sensor — HOBO Series H8; TMC6-HA and TMC6-HD probes; Onset Computer Corporation, Bourne, MA). Results showed optimal performances of RFID temperature tags, providing data with accuracy comparables with regards to the conventional methods. Notwithstanding RFID performance resulted superior at data recovery. The study of Antonucci et al. (2009) reported as a system based partially on RFID technology was used to monitor the logistics of a fresh cut vegetables product. This system was implemented to improve the logistics from the acceptance to the packaging of the product, involving four points of the supply chain: (1) product enter, (2) processing begin, (3) processing end, (4) packaging. Another application of the RFID technology was reported by Milczarek and McCarthy (2011). In their study, a method to track the peeling outcome of each individual tomato sample in a batch was carried out. This was achieved by tagging each fruit with a uniquely numbered RFID chip injected into each one through the stem scar. The end of the chip was placed into the columella chosen because it was previously observed that, if a fruit started to disintegrate during the peeling process, the most likely tissue to remain intact was the columella near the stem scar. After tagging, the fruit samples were combined with the other tomatoes from the same batch and sent through the pilot peeling unit operations. Using a portable RFID reader, the technician determined the peeling outcome for each fruit.

In the study of Chunxia et al. (2009), RFID technology adoption in vegetable supply chain (i.e., cultivating, processing, storage, transporting and retailing) was discussed. Moreover, Yang and Wang (2012) introduced the basic working principle and technical characteristics of the RFID technology, suggesting that the vegetable quality traceability could be established by this system. Then, they analyzed this current application obstacles bringing out several measurements. In addition, the use of RFID tags with probe showed to be valuable in determining the efficiency of the pre-cooling operations and low temperature abuse tracking during transportation and refrigerated storage with respect to the RFID tags without probe.

Meat Products

The implementation of RFID tracking systems in the meat production/supply chain apparently involves a greater level of technological difficulty in comparison with the horticultural sector, since monitoring is extended to live animals.

Meat quality and public safety represent an important primary public concern. Thus, legislation is progressively imposing the development of a suitable monitoring technology for meat products evaluation during production, processing, storage and distribution. For meat traceability, EU food policies are progressively imposing the implementation and use of different RFID applications. Accordingly, a series of study were published in the past few years proposing different and interesting solutions. Tomeš et al. (2009) focused on the use of RFID systems to identify cattle specimens in association with a large Biotrack database which associate each ID with whole set of biometric identifiers in order to allow the correct identification of meat sold units with the animal of origin. He also suggested that such an integrated RFID-Biotrack database system could eventually replace barcodes with the purpose of full traceability between the participants. Kong et al. (2009) developed RFID architecture in relation to meat supply chain safety control. RFID was used to identify each animal in pig farms, and to organize the information into a farm database. Luo et al. (2011) went further, by designing RFID tag for carcass, based on an online reading and writing system to be used in meat production lines within poor environment conditions (i.e., Chinese pig slaughterhouses). Authors were able not only to collect, transmit and deal with the crucial information essential for the good traceability and regarding the key processes but also printed a set of commercial meat RFID tags to be used directly on carcass and cuttings in the sales stores. As reported by Kerry et al. (2006), at present, individually RFID tagged meat products are not available to the consumer, although the use of RFID tagging of meat cuts has extended, in one case at least, to the pig processing industry from the individual pig to its primal pieces, i.e., hams. Although the purpose of this tracking scheme is for quality control, employee accountability and precision cutting, and does not extend beyond the cutting room floor or provide information about the individual animal with the final product, it does exemplify the developing use of RFID technology within the meat industry. Later, Shougang et al. (2010) developed an RFID system that worked from customers back to manufacturers, by connecting each product to the related points of interest. At sale, data from the meat products with RFID tag are entered into a database allowing the connection of the meat unit with the data flow of all other products in circulation. If necessary, this procedure facilitates the monitoring and management of product flow

reducing the time needed by the authorities for an eventual withdraw.

Dairy Products

Dairy products safety and quality have been a global public problem. An example is represented by the Sanlu milk scandal occurring in China during 2008, which raised important issues on control and management of dairy goods production (Xin and Stone 2008). As a response, the Chinese institutions promoted the use of dairy cattle radio frequency identification techniques to strengthen the supervision and management of tracing system (Tang 2009). This context was favourable to the development of different RFID control systems devoted to the traceability along the whole dairy supply chain (Zhang et al. 2010). For example, Cai and Liang (2011) implemented an RFID-based technology to trace the movement of milk packages, ensuring information flux from producers to consumers, the latter using Internet applications to obtain real-time information on the dairy item's production and processing.

RFID technology has been also used in other countries for the same purposes, and the impact of this technique is presently gaining importance. At the moment, published studies relative to the use of these technologies in Europe within the cheese industry are still few. Pérez-Aloe et al. (2007) tested different RFID applications regarding cheese traceability based on two different types of tags. Lecture trials were carried out considering markedly different condition (i.e., temperature, humidity, corrosive or saline solutions immersions or in the presence of preservative substances and oils). No significant negative effects on tags readability were reported, with the exception of those cases where metallic materials occurred in the range of the reader. In a subsequent study, Varese et al. (2008) tested the applicability of RFID technology to the dairy cheese production not only to implement the current tracking technology but also to avoid cases of imitations of Protected Designation of Origin (PDO). Two different types of small-sized tags were used and their efficiency was compared: an embedded tag directly inserted on the side of the cheese at the end of the forming process; an external tag inserted in a casein plate after the first or second turning over of the cheese. The results indicated that the positioning of the tags did not affect readability. The embedding tag proved to be more resistant during the various stages of processing, while the casein plate was more subjected to losses at handling, but only in those cases in which the cheese has a rough/uneven rind. Papetti et al. (2012) proposed the integration of an electronic tracing system with a non-destructive quality analysis for single product of a typical Italian cheese, prepared with buffalo milk. The tracing and quality information are combined on a web platform to obtain a complete

procedure defined as “infotracing system.” This system records two typologies of information (chemical and spectrophotometric) on a web platform following the product shelf-life from producer to consumer, providing web-based tools for each category. The categories involved in this system are divided into manufacturers, wholesalers, resellers, retailers and consumers who contribute separately, according to their level of membership, to provide a set of data related to each product. All collected data will enter into a centralized database.

Fishery Products

As with other perishable goods, the traceability of seafood is an important aspect to be considered during the industrial processing, in terms of safety and consistent economic aspects. These aspects directly affect the production efficacy ratio of costs/selling. Seafood represents a highly perishable food category; the stay-time of its products within different production stages and market installations directly affect their purchase probability (Deale et al. 2008; Ngai et al. 2008b; Takai and Yada 2010).

The developing global trade increases in terms of average distances that the products need to cover to reach the destination markets. Thus, the increased time needed to ship fish products abroad often generates additional problems in terms of appearance and freshness maintenance (Schröder 2008). Moreover, such difficulties can lead to potential associated diseases in a global scenario where the consumers pay more and more attention to food origin and quality (Thompson et al. 2005; Bono et al. 2010). As a result, seafood safety is presently strictly legislated, i.e., the control system made by several different compartments that are administered by different governmental organizations with variable levels of inter-coordination (Sioen et al. 2007). Both government and consumers are looking at food traceability as a means to restore confidence in the food supply and limit damages incurred by the sale and distribution of unsafe products (Thompson et al. 2005). Also, the implementation of labelling system of seafood can be also enforced in relation to authentication of origin for anti-counterfeit policies (Bono et al. 2010).

Temperature is the most important factor in prolonging the quality of seafood goods, directly influencing the production efficacy ratio (Jedermann et al. 2009). The main reason for quality losses in traded seafood items are temperature fluctuations occurring during the production stages (Jedermann et al. 2008) and affecting negatively the Cold Chain (i.e., a temperature-controlled supply chain that should be kept uninterrupted maintaining a given temperature range) (Tingman et al. 2010). As reported by Kotsianis et al. (2002), the shelf life of a product is the period of time during which quality losses do not exceed a tolerated level.

Therefore, the product's time and temperature history must be known; if not, the shelf life is uncertain (Ketzenberg and Bloemhof-Ruwaard 2009; Ruiz-Garcia and Lunadei 2010; Tingman et al. 2010).

Seafood state and quality need to be accurately controlled through the Cold Chain distribution. Therefore, the efficient management of the different steps is crucial to preserve fresh and perishables items to the market in safe and good condition (Costa et al. 2012). Under different logistic constraints, the seafood traceability must be able to provide all information about the nature, origin, and quality of commercialized products at each step of collection/production, distribution, and finally sale. In this context, there is presently an important technological effort in the implementation of “vertical” traceability systems for seafood from the field to the consumer (Costa et al. 2011a).

RFID sensors can be presently used in combination with other different categories of detectors, such as iButtons and TTIs (McCarthy et al. 2009; Liu et al. 2010). Also, RFID tags can be coupled with GPS for the mobile communication of sensors (Zhang et al. 2009). That combination allows an increase in the level of multisensory monitoring of the working environment at different industry-relevant stages and their conditions. Hazard Analysis and Critical Control Point (HACCP) challenge the benefits of RFID applications in cold chain (Aung and Chang 2010).

Bakery Products

Bakery products are also deeply subjected to physical, chemical and microbiological spoilage (Smith et al. 2004). The applications of RFID technology showed to be of help against spoilage for both processing and packaging management. Storage temperature is one of the most important factors affecting baked products preservation and quality during shipment, storage and handling. Laniel and Émond (2010) pointed out as RFID technology can be successfully used also in temperature tracking because capable of tag ability to provide data remotely and resistance to heat stress.

Generally, bakery products are temperature-sensitive and highly perishable; thus, can be vulnerable to long shipping times during sea transportation. The objective of the study of Laniel et al. (2011) was to identify the effect of RFID antenna position on tag readability of a load of frozen bread inside a sea container at two different frequencies. The results of this study showed that antenna positions have a major influence over RFID readability at specific frequencies, but readings were not dependent on antenna positioning for commercially available RFID readers. Moreover, RF propagation inside a metal environment is highly influenced by the frequency used. The tested system appears suitable for real time temperature monitoring of frozen bread inside

sea containers. This technology could be applied to other foods similar to frozen bread.

Beverages

Bernardi et al. (2008) reported a case study evaluating the implementation of RFID to an anti-counterfeit mechanism in selected wine production environments. The work introduced the use of a personal digital assistant (PDA) with a public/private key mechanism involving both, passive RFID internal memory and unique RFID identifier to allow the reseller and the final user to verify whether the bottle is original.

Another application developed is represented by the detection of the liquid level in both commercial beverage glasses and bottles, which is an important service to this industry in order to obtain a standardized product and good approval ratings (Bhattacharyya et al. 2010). Nowadays, the sensing approaches rely either on visual inspection or on sensor electronics to verify liquid level. In such a work, the RFID tag antenna was used as a low-cost alternative in the service industry, detecting the volume of liquid in a beverage glass by mapping a change in received signal strength indication (RSSI) power.

Other Food Products

As reported by Singh et al. (2007), a data recorder with RFID features was obtained for a quality control application in the shell-egg industry. Among all the agricultural commodities that are subjected to the possibility of damage during the course of production, shell-eggs are particularly susceptible to being cracked or broken during the production operation. An instrumented egg was designed to be placed anywhere in the egg gathering, conveying or packaging systems travelling amongst the real eggs through the production process, identifying abuse points and reporting location and magnitude of abuse instantly to the user in real time. If the data recorder dropped, rolled, or came into contact with a solid object, it sent a reading to a handheld computer; the egg also transmitted a temperature reading and flagged high-pressure areas.

Conclusions

This review describes how the widening use of RFID technology for the traceability of agri-food products takes place within the different stages of the supply chain—from production, through distribution, up to the final merchandizing of the products. The reasons for RFID's versatility and logistic advancements are attributed to the possibility of coupling a variegated typology of radio frequency smart

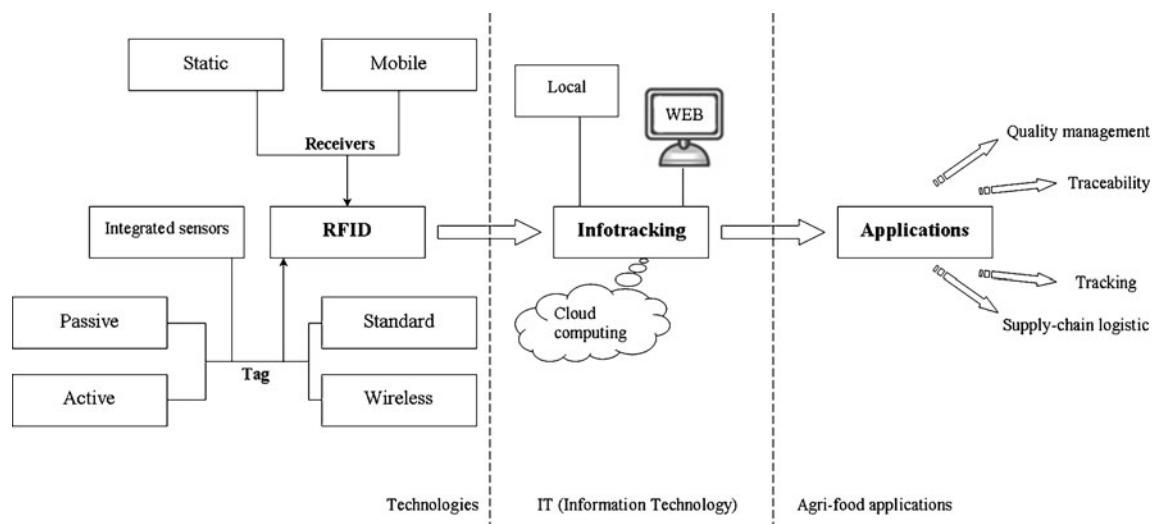


Fig. 3 Scheme summarizing the technological devices, the information technology systems and the agri-food applications reviewed in this study

tags to different environmental sensors through the new tools provided by the fast developing WSN technology.

RFID tools are advanced technological devices which allow the following: (a) improvement of the vertical traceability of products and the monitoring of the status of their processing conditions at each stage of the production/distribution chain; (b) efficiency enhancement of processes; (c) strict quality monitoring, giving handlers the opportunity to assess the precise permanence of goods at each stage of the distribution/supply chain, which in turn may cause important issues in relation to public health when goods are highly prone to decay; (d) improvement on information accuracy for government agencies and consumers/customs officers, which can track in real-time backward the provenience and conditions of acquired goods.

As referenced RFID implementations in the agri-food sector are increasing at a great rate, technological advances are also following the applicability opportunities. However, real applications of RFID technologies are still limited because of various technical and economical obstacles. From a management point of view, for example, there are scarce applications on infotracking systems including the possibilities to explore new web-based Cloud Computing, which refers to the capability to upload data (i.e., tag readings) into web applications that are delivered as services over the Internet and in which managing hardware and systems software are located in different data centers (i.e., the Cloud) that deliver those services worldwide. In the research development scenario described by this review, we propose the use of RFID technology and associated infotracking in a series of potential applications, which are summarized in Fig. 3.

Finally, considering these applications, it can be affirmed that although RFID technology presents mainly positive

effects, it also has some drawbacks. The valuable opportunities to couple cloud computing to RFID tracking technology are: (a) a reduction in the costs of industrial monitoring technology; (b) informative integration; (c) optimization of intra- and inter-company logistics (best efficiencies and cost reductions) in relation to quality preservation and safety implementation. Conversely, the constraints to the widening use of RFID technology seems to reside in the adaptation of its designs to the elevated complexity and plurality of different supply chains, managed by professionals from very different backgrounds, which prevents the organic structuring of this technology.

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